

Track: EdTech | Challenge ID: ET-07

Design a Gamified Learning Engagement Platform with Adaptive Challenges, Scoring, Badges, Leaderboards, and Participation Analytics

Problem Description

Gamification can increase participation when game mechanics are aligned with learning tasks, but simple point accumulation may reward activity without reflecting meaningful progress. The challenge is to develop a learning-engagement platform that combines quizzes or challenges with transparent scoring, badges, progression rules, and analytics. The system should include at least one adaptive or mastery-aware element so that engagement features respond to learner performance rather than functioning as a static leaderboard.

Technical Scope and Minimum Requirements

The prototype should support learner accounts, challenge/quiz content, scoring, levels or badges, leaderboard logic, participation history, and an instructor/admin view. Teams should implement at least one adaptive mechanism such as difficulty adjustment, targeted challenge recommendation, streak recovery, or mastery-based badge conditions. Rules for points and badges should be documented and resistant to obvious repeated-action abuse. Dashboard analytics should distinguish participation from achievement.

Expected Outcomes

A successful system will provide an engaging learner workflow while giving instructors useful visibility into participation and performance. Judges should be able to complete challenges, earn points or badges, observe leaderboard changes, and see adaptive behaviour after strong or weak performance. Expected outputs include score history, badge criteria, progression status, topic performance, participation frequency, and class-level engagement indicators.

12-Hour Minimum Deliverable and Live Demonstration

Within 12 hours, teams should deliver a playable prototype with a prepared set of learning challenges and multiple synthetic users. The live demo must show scoring, at least one badge or level transition, leaderboard update, adaptive challenge behaviour, and instructor analytics. Complex multiplayer gaming, payment systems, or large-scale LMS integration are outside the minimum scope.